

Topic:

Intent

⇒ The dictionary meaning of Intent is intention or purpose, thus here intent defined as the intention to do any action.

⇒ Intent can also be said as message, to perform an action on the screen.

⇒ Actions that can be performed by an intent are:

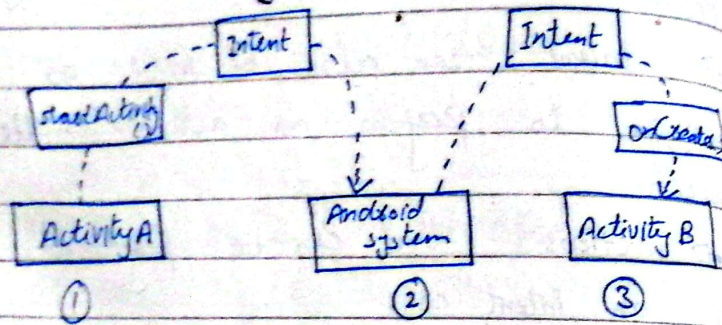
- Launch another activity
- Dial a phone call
- Displaying the list of contacts
- Display a web page
- Start the service
- Broadcast a message etc.

⇒ Intent are also used to communicate b/w App Components and it can also provide the connectivity b/w two apps.

⇒ Programmatically Intent are used as objects which is used to pass the information b/w activities within single or multiple applications.

⇒ for example:

```
Intent intent = new Intent(...);  
startActivity(intent); or startService(...);
```



⇒ Intents are used as a message passing mechanism that let to declare intention that an action be performed, usually with a particular piece of data.

Uses of Intents

⇒ To support interaction b/w any of the App components available on android studio.

- To start new activity implicitly or explicitly.
- To broadcast message across the system.

◦ The most common uses of intents to bind application components. Intents are used to start, stop and transition b/w the Activities within an application.

Types of Intents

⇒ There are two types of intents:

- Implicit intent
- Explicit intent

⇒ Implicit Intent:

This intent specifies an action that can be invoked by an app on the device which enables us to perform an action. It does not have the exact knowledge about the landing Component.

It can open another app or

- it only specifies action to be performed. Then android os will fill out components.
- uri used to specify data types. that will respond to and action.
- its own app's component exist.

e.g

Downloading song, PDF, image, document, dial, call, map location etc.

⇒ It does not specify the component directly.
Syntax of open Google map:

```
Intent intent = new Intent ( Intent.ACTION_VIEW
intent.setData (Uri.parse ("http://www.google.
startActivity(intent); com"));
```

Example 2:

To dial a number:

```
Intent i = new Intent ( Intent.ACTION_DIAL);
i.setData (Uri.parse ("tel:555-2368"));
startActivity(i);
```

(2) Explicit Activity:

⇒ This intent specifies the component in an app. It is one that we use to launch a specific app component, such as particular activity or service in an application.

⇒ Using this intent we can pass data from one activity to another activity.

⇒ To explicitly select an Activity class to start, create a new Intent specifying the current application context and the class of the Activity to launch. Pass this intent in to start activity.

Code:

```
Intent intent = new Intent ( myActivity.this,
MyOtherActivity.class);
startActivity(intent);
```

⇒ After calling startActivity, the new Activity will be created and

become visible and active, moving to the top of the stack.

⇒ Calling `finish()` programmatically on the new Activity will close it and remove it from the stack. Alternatively, users can navigate to the previous Activity using the device's Back button.

Q. Write code to send data from one activity to other activity using Explicit Intent?

```
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class SenderActivity extends AppCompatActivity
```

```
{
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState)
```

```
{
```

```
super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_sender);
```

```

Button sendButton = findViewById(R.id.  

sendButton.setOnClickListener(new View.  

OnClickListener()

```

```

{
    @Override
    public void onClick (View view)
    {

```

// Create an Intent to send data to the receiver activity

```

Intent i = new Intent (SenderActivity.this,  

ReceiverActivity.class);

```

// put data into the intent using a key-value pair

```

intent.putExtra ("message", "Hello  

from Sender Activity");

```

// Start the receiver activity

```

startActivity(i);
}
}); }

```

Receiver Activity

```

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.  

AppCompatActivity

```

```

Public class ReceiverActivity extends  

AppCompatActivity

```

```

{
    @Override
    protected void onCreate (Bundle  

savedInstanceState)

```

```

{
    super.onCreate (savedInstanceState);
    setContentView(R.layout.activity_receiver);

```

// Get the Intent that started this activity

```

Intent intent = getIntent();

```

// Retrieve the data from the Intent using the key

```
String message = intent.getStringExtra("message")
```

// Display the received data in a TextView

```
TextView textView = findViewById(R.id.message_textview);  
textView.setText(message);  
}  
}
```